

# **Adapting a Sub-1B Vision-Language Model for Document Understanding — A One-Factor-at-a-Time LoRA Ablation on the Unified Document Dataset (UDD)**

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Model: LiquidAI/LFM2.5-VL-1.6B (LoRA)

Environment: Kubeflow thor2-h100 (NVIDIA H100 NVL)

Dataset: danielcsb/UDD (public Hub release) — 9086 images / 16708 QAs / 32 source benchmarks / 7 tasks

Regime: 512px images, 300 steps/arm, single seed. Relative signal, NOT absolute SOTA.

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# 1. Executive summary

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Question: when LoRA-finetuning a small ( $\leq 1B$ ) document VLM, WHICH training factor actually helps? We vary one factor at a time across 24 arms (A1..A6), train each, and score the base-vs-adapted model. Two evaluation designs:

- **v1** = 4 suites, 3 of them EXTERNAL synthetic probes + UDD heldout (64-cap).
- **v2** = UDD-ONLY: per-task heldout + localization sliced by document quadrant (region).

**HEADLINE:** 'HOW you train' (A5 LoRA placement, A6 rank) beats 'WHAT signal you add' (A1 spotting, A2 rationale). Best recipe: LoRA on LLM attn/mlp, rank 16-32, spot+reason data.

**KEY v2 FINDING (region):** LLM-module placements lift vqa/reasoning but SYSTEMATICALLY degrade localization (worst top-left, llm\_mlp -0.10); only spotting arms keep grounding.

**REVERSAL:** A2 chain-of-thought/rationale supervision did NOT help — it hurt.

## 2. Experiment design (per factor: WHAT / VERIFY / HOW / CONCLUSION)

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### 2.1 A1 Spotting

- **WHAT (variation):** Add bbox('where') targets to training or not (on=vqa+kie+localization thirds vs off=vqa+kie halves), equal N=796, same steps.
- **VERIFY (hypothesis):** Does teaching location improve extraction/space vs answer-only? Does it transfer beyond localization?
- **HOW (method):** Train A1\_spotting\_on vs A1\_spotting\_off, score UDD heldout before/after.
- **CONCLUSION:** Both  $\sim +0.1$  heldout; on keeps grounding neutral but trades vqa. Spotting's cross-transfer NOT confirmed at this scale.

### 2.2 A2 Reasoning

- **WHAT (variation):** Same (image,question,element) with target = rationale->answer (chain) vs answer-only. Only the rationale text differs. N=2327.
- **VERIFY (hypothesis):** Does chain-of-thought supervision improve integrative reasoning, or is answer-only enough?
- **HOW (method):** Train A2\_reason\_chain vs A2\_reason\_answer, controlled pair.
- **CONCLUSION:** REVERSAL: both  $\leq 0$  on heldout; rationale not better than answer. CoT supervision hurt at this scale/quality.

### 2.3 A3 Combination

- **WHAT (variation):** {base=vqa+kie} / {+spot} / {+reason} / {+spot+reason}, equal N=796.
- **VERIFY (hypothesis):** Is it the act of ADDING signals, or does plain task combination already capture it? Are spot & reason complementary?
- **HOW (method):** Train 4 arms same total, compare heldout per task.

- **CONCLUSION:** spot\_reason strong on vqa (+0.29) & reasoning; base already strong. Combination  $\geq$  single.

## 2.4 A4 Multilingual

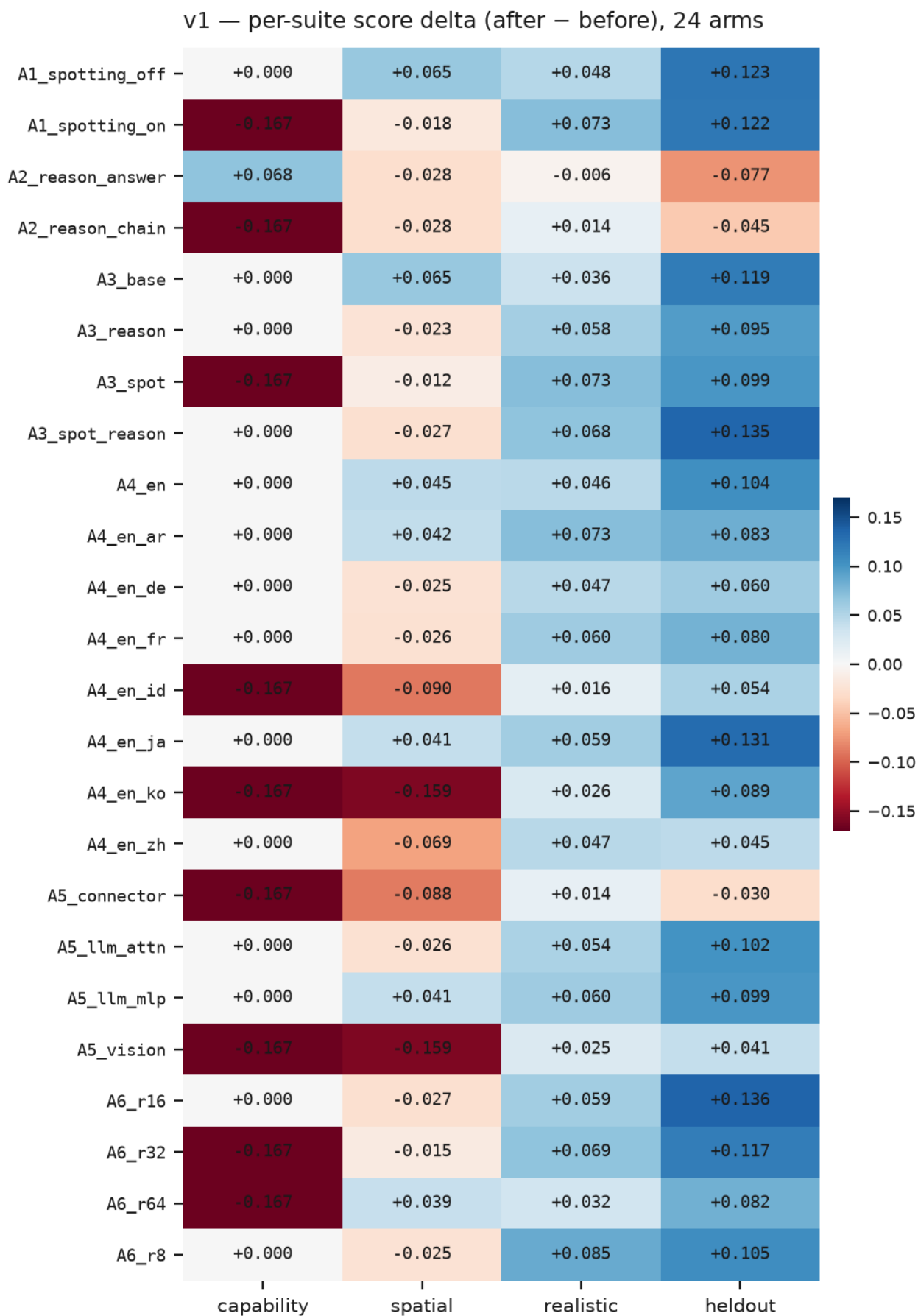
- **WHAT (variation):** Training language mix {en} vs {en+X}, X in {ko,ja,ar,zh,id,fr,de}, equal N=178.
- **VERIFY (hypothesis):** Does multilingual mixing beat single-language? Which language pairs transfer/interfere?
- **HOW (method):** Train per mix, score UDD heldout (+ per-language).
- **CONCLUSION:** en alone best overall (reasoning +0.25); adding languages diluted; distant scripts (ko/id/zh) weakest.

## 2.5 A5 Placement

- **WHAT (variation):** Fixed composite mix; vary WHERE LoRA is placed: vision / connector / llm\_attn / llm\_mlp.
- **VERIFY (hypothesis):** Which module routes which capability?
- **HOW (method):** Data & rank fixed, placement varied; read per-task + per-region transfer.
- **CONCLUSION:** llm\_mlp best (vqa+reasoning); connector worst. NEW: LLM-module placements degrade localization (see Section 5.2).

## 2.6 A6 HPO (rank)

- **WHAT (variation):** Fixed mix & placement; vary LoRA rank r in {8,16,32,64} ( $\alpha=2r$ ).
- **VERIFY (hypothesis):** Optimal rank? Does too-high rank overfit small data?
- **HOW (method):** Vary rank only, score heldout + overfit signals.
- **CONCLUSION:** Moderate rank r16-r32 best (r32 vqa +0.54); r8 weakest; r64 overfits (v1 realistic:consistency -0.75).



### 3. The data (UDD): 32 benchmarks and their internal differences

UDD unifies 32 public benchmarks (one row per image, native QA lists, leakage-safe fold). Per-source breakdown:

| source         | imgs | QAs  | task           | metric      | med_dim   | held |
|----------------|------|------|----------------|-------------|-----------|------|
| ai2d           | 300  | 300  | vqa            | exact       | 521x404   | 24   |
| charxiv        | 300  | 300  | reasoning      | exact       | 1000x660  | 26   |
| doclaynet      | 300  | 300  | localization   | grounding   | 1000x1000 | 28   |
| docvqa         | 300  | 300  | vqa            | anls        | 776x1000  | 23   |
| dvqa           | 300  | 1500 | reasoning      | anls        | 448x448   | 30   |
| iam            | 300  | 300  | recognition    | ned         | 1000x65   | 30   |
| im2latex       | 300  | 300  | recognition    | ned         | 289x56    | 26   |
| infovqa        | 300  | 300  | vqa            | anls        | 417x1000  | 16   |
| latexocr       | 300  | 300  | recognition    | ned         | 240x40    | 27   |
| mathvista      | 300  | 300  | reasoning      | exact       | 448x369   | 34   |
| ocrbench       | 300  | 300  | vqa            | ocrbench    | 184x67    | 28   |
| ocrvqa         | 300  | 1417 | vqa            | exact       | 333x500   | 33   |
| plotqa         | 300  | 1500 | reasoning      | anls        | 1000x604  | 40   |
| publaynet      | 300  | 300  | localization   | grounding   | 612x792   | 26   |
| pubtabnet      | 300  | 300  | table          | anls        | 503x161   | 37   |
| sroie          | 300  | 300  | kie            | ned         | 78x19     | 37   |
| stvqa          | 300  | 394  | vqa            | anls        | 500x333   | 30   |
| synthdog_en    | 300  | 300  | recognition    | ned         | 897x856   | 27   |
| synthdog_ko    | 300  | 300  | recognition    | ned         | 905x871   | 25   |
| tatqa          | 300  | 1499 | reasoning      | anls        | 699x226   | 26   |
| textvqa        | 300  | 300  | vqa            | exact       | 1000x750  | 38   |
| mtvqa          | 299  | 990  | vqa            | anls        | 1000x750  | 32   |
| ocrbench_v2    | 299  | 301  | vqa            | exact       | 562x1000  | 33   |
| omnidocbench   | 299  | 299  | recognition    | ned         | 708x1000  | 27   |
| rvl_cdip       | 298  | 298  | classification | exact       | 754x1000  | 29   |
| chartqa        | 296  | 300  | reasoning      | relaxed_acc | 850x600   | 35   |
| screenqa       | 294  | 298  | vqa            | anls        | 562x1000  | 33   |
| docmatix       | 287  | 1441 | vqa            | anls        | 773x1000  | 21   |
| hallusionbench | 286  | 574  | reasoning      | exact       | 696x502   | 24   |
| visualmrc      | 278  | 947  | vqa            | anls        | 571x1000  | 30   |
| cord           | 100  | 100  | kie            | anls        | 667x1000  | 11   |
| funsd          | 50   | 50   | kie            | ned         | 754x1000  | 4    |

## SIX AXES OF DIFFERENCE WITHIN UDD:

- **(a) 7 output contracts:** span-answer / gold-list(ANLS) / transcription(NED) / structured-record(F1) / structure-tree(TEDS) / layout(grounding) / reliability. Same 'doc understanding' label, different heads.
- **(b) Image geometry extremes:** crops (sroie 78x19, latexocr 240x40, iam 1000x65 single-line) vs full-page (doclaynet 1000x1000, textvqa 1000x750). 512px cap shrinks ONLY full-page sources.
- **(c) QA density:** most 1 QA/img; chart/doc-VQA multi-QA (dvqa/plotqa/tatqa/ocrvqa/docmatix ~5 QA/img).
- **(d) Language:** en 83.6%; thin multilingual tail (ja 60, fr 49, de 44, ar 42) -> A4 equal-N shrinks to 178.
- **(e) Metric tolerance** varies even within a task (vqa: exact vs anls vs ocrbench).
- **(f) Structural density:** kie n\_fields median 40 (max 433); localization n\_regions median 11 (max 66).

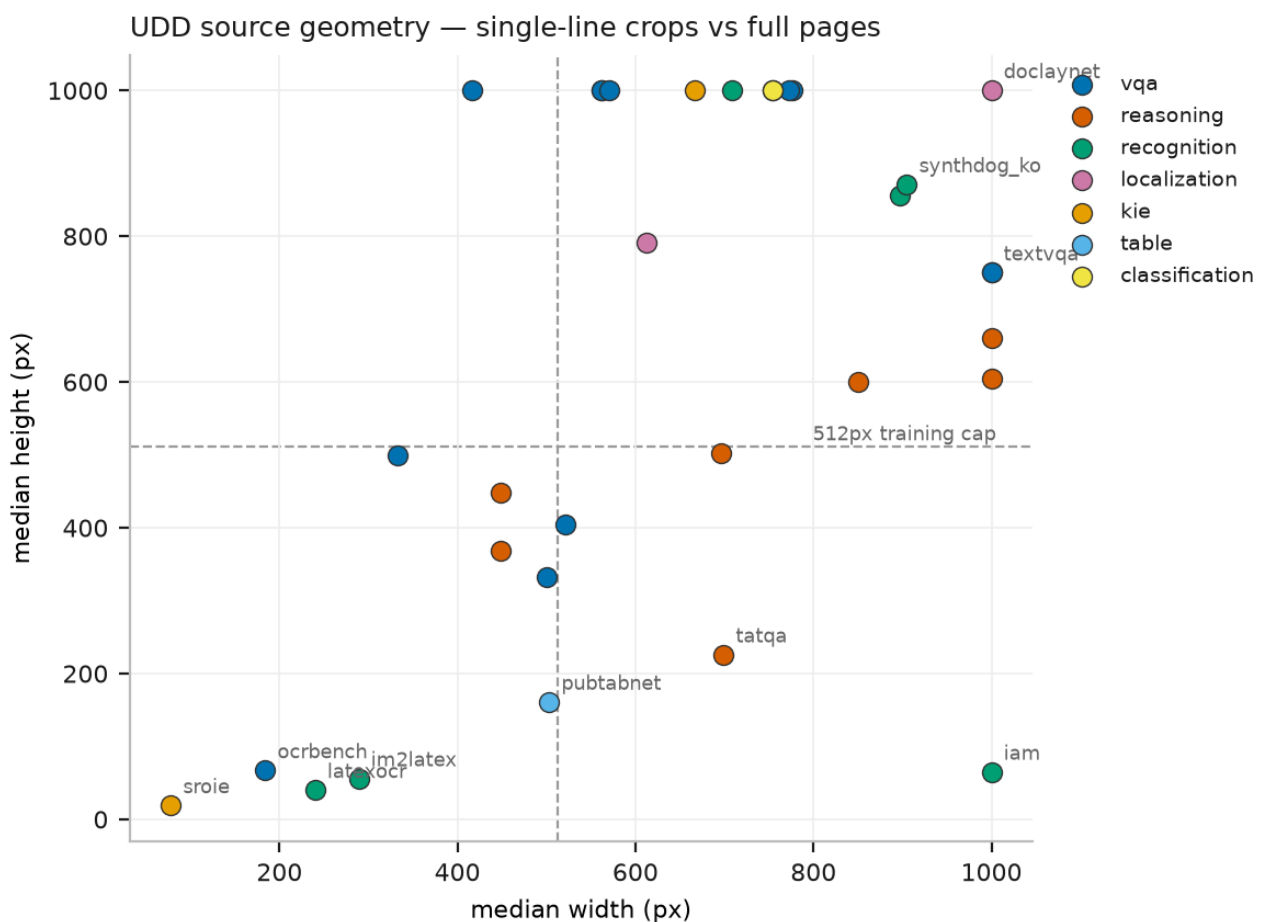


Figure 2 — image-geometry extremes across the 32 sources.

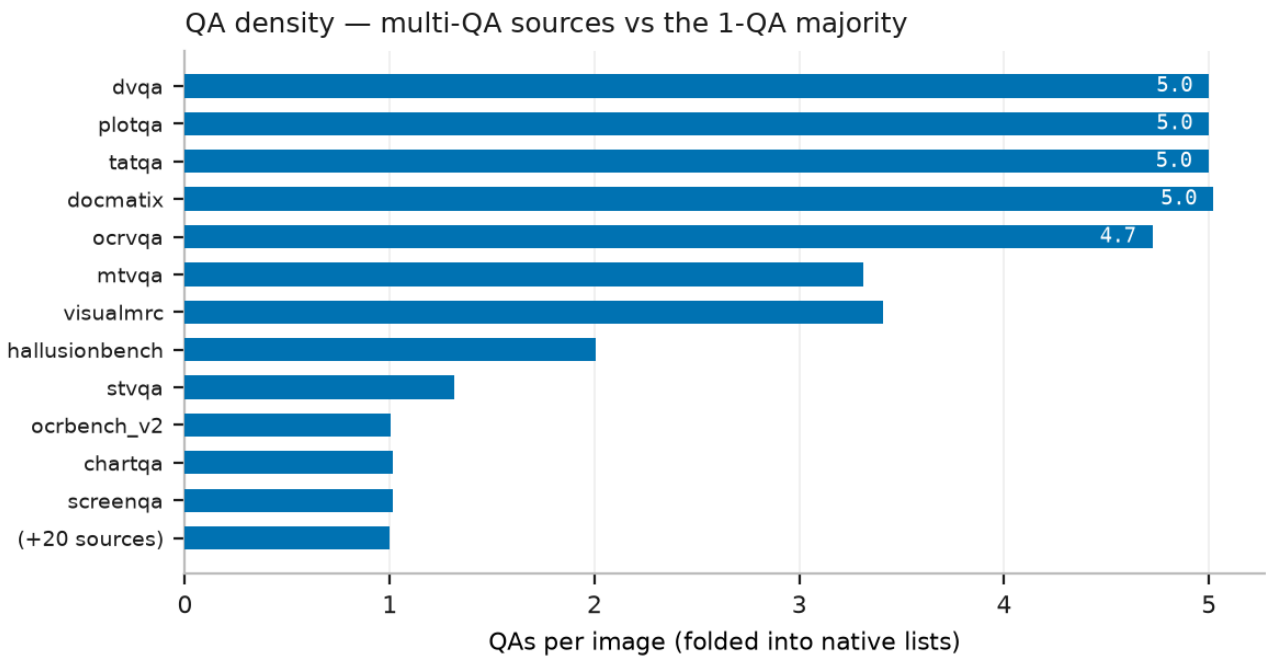


Figure 3 — QA density: multi-QA sources vs the 1-QA majority.

## 4. v1 results (4 suites incl. external probes; before->after delta)

| arm              | capability          | spatial             | realistic           | heldout             |
|------------------|---------------------|---------------------|---------------------|---------------------|
| A1_spotting_off  | 0.833→0.833(+0.000) | 0.694→0.759(+0.065) | 0.264→0.312(+0.048) | 0.284→0.407(+0.123) |
| A1_spotting_on   | 0.833→0.667(-0.167) | 0.694→0.676(-0.018) | 0.264→0.337(+0.073) | 0.284→0.406(+0.122) |
| A2_reason_answer | 0.833→0.901(+0.068) | 0.694→0.667(-0.028) | 0.264→0.259(-0.006) | 0.284→0.207(-0.077) |
| A2_reason_chain  | 0.833→0.667(-0.167) | 0.694→0.667(-0.028) | 0.264→0.278(+0.014) | 0.284→0.239(-0.045) |
| A3_base          | 0.833→0.833(+0.000) | 0.694→0.759(+0.065) | 0.264→0.300(+0.036) | 0.284→0.403(+0.119) |
| A3_reason        | 0.833→0.833(+0.000) | 0.694→0.672(-0.023) | 0.264→0.322(+0.058) | 0.284→0.379(+0.095) |
| A3_spot          | 0.833→0.667(-0.167) | 0.694→0.682(-0.012) | 0.264→0.337(+0.073) | 0.284→0.383(+0.099) |
| A3_spot_reason   | 0.833→0.833(+0.000) | 0.694→0.667(-0.027) | 0.264→0.332(+0.068) | 0.284→0.419(+0.135) |
| A4_en            | 0.833→0.833(+0.000) | 0.694→0.739(+0.045) | 0.264→0.310(+0.046) | 0.284→0.388(+0.104) |
| A4_en_ar         | 0.833→0.833(+0.000) | 0.694→0.736(+0.042) | 0.264→0.337(+0.073) | 0.284→0.367(+0.083) |
| A4_en_de         | 0.833→0.833(+0.000) | 0.694→0.669(-0.025) | 0.264→0.311(+0.047) | 0.284→0.344(+0.060) |
| A4_en_fr         | 0.833→0.833(+0.000) | 0.694→0.669(-0.026) | 0.264→0.324(+0.060) | 0.284→0.364(+0.080) |
| A4_en_id         | 0.833→0.667(-0.167) | 0.694→0.605(-0.090) | 0.264→0.280(+0.016) | 0.284→0.338(+0.054) |
| A4_en_ja         | 0.833→0.833(+0.000) | 0.694→0.735(+0.041) | 0.264→0.323(+0.059) | 0.284→0.415(+0.131) |
| A4_en_ko         | 0.833→0.667(-0.167) | 0.694→0.535(-0.159) | 0.264→0.290(+0.026) | 0.284→0.373(+0.089) |
| A4_en_zh         | 0.833→0.833(+0.000) | 0.694→0.626(-0.069) | 0.264→0.311(+0.047) | 0.284→0.329(+0.045) |
| A5_connector     | 0.833→0.667(-0.167) | 0.694→0.606(-0.088) | 0.264→0.278(+0.014) | 0.284→0.254(-0.030) |
| A5_llm_attn      | 0.833→0.833(+0.000) | 0.694→0.668(-0.026) | 0.264→0.318(+0.054) | 0.284→0.386(+0.102) |
| A5_llm_mlp       | 0.833→0.833(+0.000) | 0.694→0.735(+0.041) | 0.264→0.324(+0.060) | 0.284→0.383(+0.099) |
| A5_vision        | 0.833→0.667(-0.167) | 0.694→0.535(-0.159) | 0.264→0.289(+0.025) | 0.284→0.325(+0.041) |



| arm    | capability          | spatial             | realistic           | heldout             |
|--------|---------------------|---------------------|---------------------|---------------------|
| A6_r16 | 0.833→0.833(+0.000) | 0.694→0.667(-0.027) | 0.264→0.324(+0.059) | 0.284→0.420(+0.136) |
| A6_r32 | 0.833→0.667(-0.167) | 0.694→0.679(-0.015) | 0.264→0.333(+0.069) | 0.284→0.401(+0.117) |
| A6_r64 | 0.833→0.667(-0.167) | 0.694→0.733(+0.039) | 0.264→0.296(+0.032) | 0.284→0.366(+0.082) |
| A6_r8  | 0.833→0.833(+0.000) | 0.694→0.669(-0.025) | 0.264→0.349(+0.085) | 0.284→0.389(+0.105) |

NOTE: capability n=6 (+0.167 noise). 3 of 4 suites are NOT UDD -> conflates improvement with cross-dataset transfer.

## 5. v2 results (within-UDD, external probes removed)

### 5.1 Task transfer — delta per UDD task (heldout only, ~24/task)

| arm              | vqa   | reason | kie   | recogn | table | classi | ALL   |
|------------------|-------|--------|-------|--------|-------|--------|-------|
| A1_spotting_off  | +0.46 | +0.25  | +0.08 | -0.01  | +0.00 | +0.00  | +0.10 |
| A1_spotting_on   | +0.25 | +0.21  | -0.00 | -0.01  | +0.00 | +0.04  | +0.06 |
| A2_reason_answer | +0.00 | -0.08  | -0.00 | +0.00  | +0.00 | +0.00  | -0.03 |
| A2_reason_chain  | +0.00 | +0.00  | -0.00 | -0.00  | +0.00 | +0.00  | -0.00 |
| A3_base          | +0.50 | +0.25  | +0.09 | +0.00  | +0.00 | +0.00  | +0.10 |
| A3_reason        | +0.33 | +0.25  | +0.02 | -0.00  | +0.00 | +0.00  | +0.07 |
| A3_spot          | +0.21 | +0.17  | +0.00 | -0.01  | +0.00 | +0.04  | +0.05 |
| A3_spot_reason   | +0.29 | +0.17  | +0.04 | -0.01  | +0.00 | +0.04  | +0.06 |
| A4_en            | +0.00 | +0.25  | -0.00 | +0.00  | +0.00 | +0.12  | +0.04 |
| A4_en_ar         | +0.00 | +0.17  | -0.00 | +0.00  | +0.00 | +0.00  | +0.01 |
| A4_en_de         | +0.00 | +0.21  | -0.00 | +0.00  | +0.00 | +0.00  | +0.02 |
| A4_en_fr         | +0.00 | +0.21  | -0.00 | +0.00  | +0.00 | +0.04  | +0.03 |
| A4_en_id         | +0.00 | +0.12  | +0.03 | +0.00  | +0.00 | +0.04  | +0.01 |
| A4_en_ja         | +0.00 | +0.21  | -0.00 | +0.00  | +0.00 | +0.04  | +0.03 |
| A4_en_ko         | +0.00 | +0.17  | -0.00 | -0.00  | +0.00 | +0.00  | +0.02 |
| A4_en_zh         | +0.00 | +0.12  | -0.00 | +0.00  | +0.00 | +0.00  | +0.01 |
| A5_connector     | +0.00 | +0.04  | -0.00 | -0.00  | +0.00 | +0.00  | -0.00 |
| A5_llm_attn      | +0.00 | +0.12  | +0.03 | -0.01  | +0.00 | +0.04  | +0.01 |
| A5_llm_mlp       | +0.21 | +0.17  | +0.04 | -0.00  | +0.00 | +0.04  | +0.04 |
| A5_vision        | +0.00 | +0.12  | -0.00 | +0.00  | +0.00 | +0.00  | +0.01 |
| A6_r16           | +0.33 | +0.17  | +0.02 | -0.01  | +0.00 | +0.04  | +0.06 |
| A6_r32           | +0.54 | +0.25  | -0.00 | -0.01  | +0.00 | +0.00  | +0.09 |
| A6_r64           | +0.33 | +0.17  | +0.07 | -0.00  | +0.00 | +0.04  | +0.06 |
| A6_r8            | +0.00 | +0.25  | +0.03 | -0.01  | +0.00 | +0.08  | +0.03 |

## 5.2 Localization region transfer — delta grounding by document quadrant (KEY NEW FINDING)

| arm             | TL    | TR    | BL    | BR    |
|-----------------|-------|-------|-------|-------|
| A1_spotting_off | -0.02 | -0.01 | -0.02 | +0.02 |
| A1_spotting_on  | +0.00 | +0.00 | -0.00 | +0.00 |
| A3_base         | -0.02 | +0.01 | -0.02 | +0.02 |
| A3_reason       | -0.04 | +0.01 | -0.01 | -0.00 |
| A3_spot         | +0.00 | +0.01 | +0.00 | +0.01 |
| A3_spot_reason  | -0.01 | -0.02 | -0.01 | -0.02 |
| A5_connector    | -0.07 | -0.01 | -0.02 | +0.02 |
| A5_llm_attn     | -0.05 | -0.06 | -0.02 | -0.03 |
| A5_llm_mlp      | -0.10 | -0.05 | -0.04 | -0.03 |
| A5_vision       | -0.08 | -0.04 | -0.04 | -0.01 |

**BOXED KEY FINDING:** A5 LLM-module placements (llm\_mlp/attn/vision) systematically degrade localization, worst top-left (llm\_mlp TL -0.10, vision -0.08, connector -0.07). Only spotting arms (A1\_spotting\_on, A3\_spot) keep grounding neutral. => trade-off: reading/reasoning via LLM modules is bought at the cost of locating. Top-left is the most fragile region across all arms.

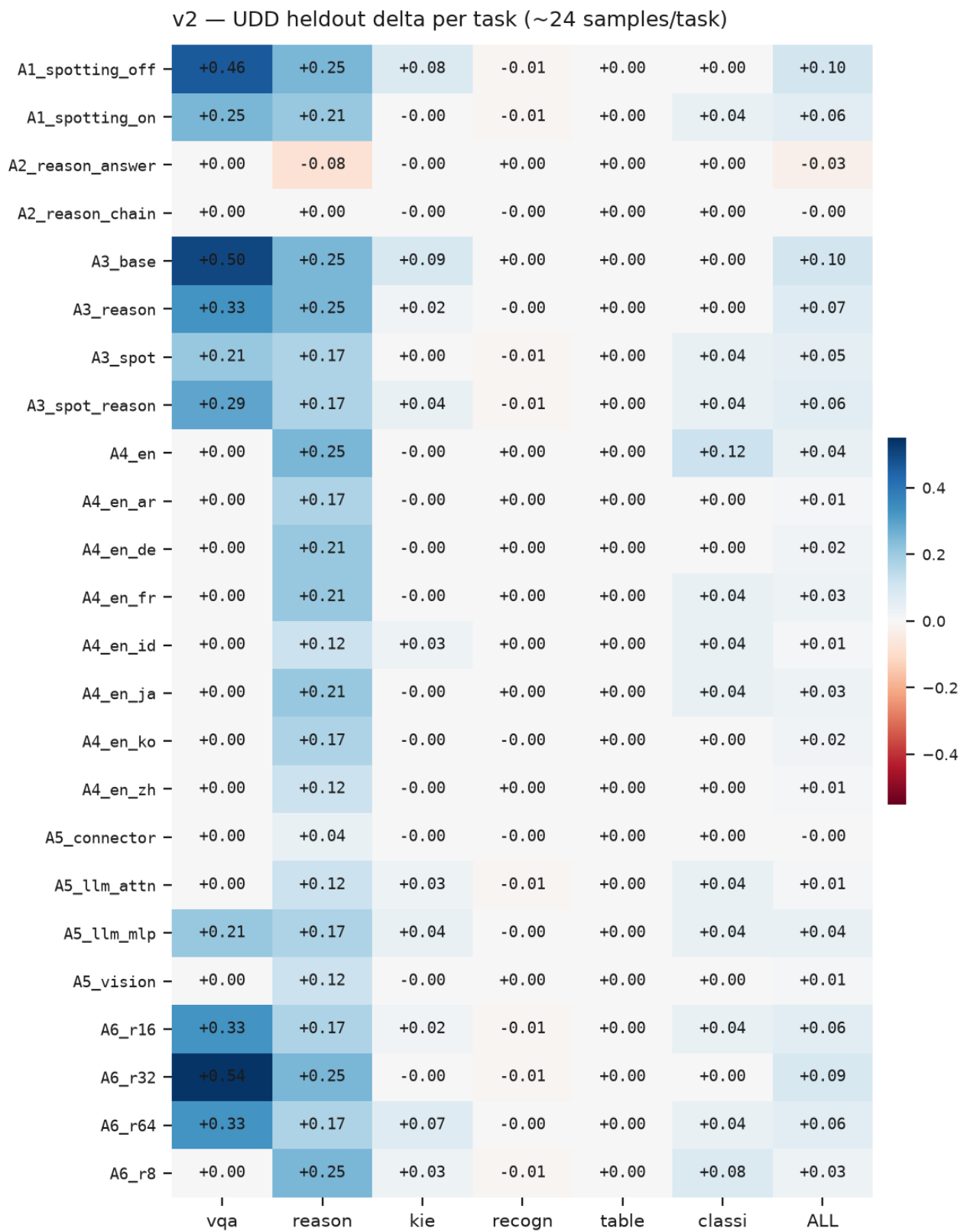


Figure 4 — v2 per-task heldout delta.

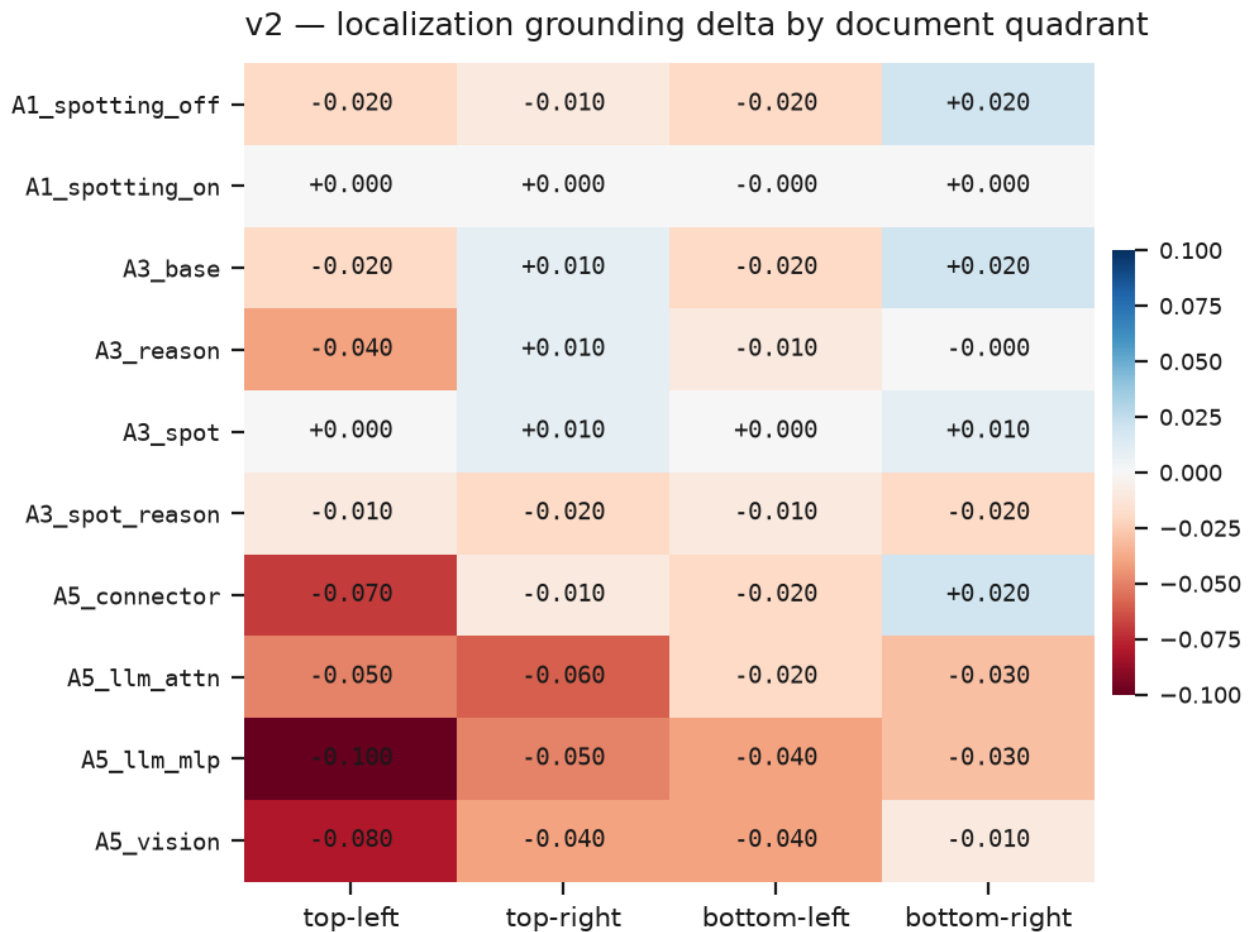


Figure 5 — v2 grounding delta by document quadrant.

## 6. Conclusions

1. HOW-you-train (A5 placement, A6 rank) > WHAT-signal (A1 spotting, A2 rationale) at this scale.
2. Recommended recipe: LoRA on LLM attn/mlp, rank 16-32, data mix incl. spot+reason.
3. A2 rationale/chain-of-thought supervision was a REVERSAL — it hurt.
4. Placement routes capability AND trades it: LLM modules -> vqa/reasoning up, localization down. Keep spotting targets in the data if grounding matters.
5. Evaluation design matters: external-probe v1 mixed transfer-to-another-dataset with improvement; within-UDD task+region (v2) is the clean read and revealed the region trade-off v1 could not.

## 7. Caveats

- Relative signal, not absolute conclusions: 512px (not native res), 300 steps, single seed, small LoRA.
- Hub UDD release (9,086 imgs,  $\leq 300$ /source) is smaller than the author's full build (28,299).
- v1 capability suite  $n=6$  (+0.167 per sample) -> treat as noise.
- v2 heldout is a balanced 192-row subset (task  $\sim 24$ , localization region  $\sim 12$ ) -> directional, not high-precision.

- recognition/table/classification deltas ~flat: too few samples / 512px hurts OCR crops.